# Robotic Applications with Particle Swarm Optimization (PSO)

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***Abstract-*** Swarm Intelligence (SI) is inspired from the study of colonies of swarms in Nature. This paper introduces a study originated on swarm intelligence, Particle Swarm Optimization (PSO). Three case studies- on one link arm, a SCARA robot and a electrohydraulic robot- are presented on both simulation and experimental base to show effectiveness and simplicity of application. Convergence results of the cases are shown on robot trajectory control (PSO-PID) and robotic system identification (PSO-NN).

***Keywords-*** *Particle swarm optimization, PID control, Neural Networks, System identification, SCARA Robot, one link arm, electrohydraulic robot*

## 1. INTRODUCTION

Nowadays, researchers are utilizing different soft computing techniques more and more to improve the performance of controllers in robotic manipulators. Methods including neural networks, fuzzy logic and evolutionary algorithms are among the most popular ones. In addition to these techniques, a relatively new population based optimization technique named Particle Swarm Optimization (PSO) has attracted the researcher's interest. The algorithm could be situated among the so called soft computing techniques. PSO algorithm, which is generally preferred because of its easy implementation, stable convergence characteristic and good computational efficiency is a Swarm Intelligence (SI) algorithm. PSO algorithm can also be combined with neural networks and fuzzy logic control systems.

As a soft computing technique, PSO shares many similarities with Genetic Algorithms (GAs). The system is initialized with a population of random solutions and searches for optima by updating generations using a defined fitness value. However, unlike GAs, PSO has no evolution operators such as crossover and mutation. In PSO, the potential solutions, called particles, fly through the problem space by following the current optimum particles and information sharing is done with the help of best particle. Another advantage of PSO compared to GAs is that it is easier to implement PSO and there are few parameters to adjust.

In this paper a brief overview of swarm intelligence and PSO are presented together with three case studies. In each case, the control and identification problem is introduced with PSO. The performance of PSO is analyzed on different systems; one link arm, a Selectively Compliant Articulated Robot Arm (SCARA) and a electro hydraulic robot. In the first and the second case simulation and experimental results are presented for PSO-PID alternative whereas in the third case PSO-NN is used for the system identification.

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### 1.1 Swarm Intelligence (SI) and Particle Swarm Optimization (PSO) Algorithm

The particle swarm concept originated as a simulation of simplified social system. The original intent was to graphically simulate the choreography of bird of a bird block or fish school. However, it was found that particle swarm model can be used as an optimizer.[1-4]. In 1995, it is found that the algorithm introduced by Dr. Eberhart and Dr. Kennedy [5] can be applied to solve various optimization problems and problems that can be converted to function optimization problems. In analogy with evolutionary computation paradigms, a swarm is similar to a population, while a particle is similar to an individual [4]. The algorithm includes a swarm of particles moving in the n-dimensional problem space where each particle represents a potential solution having a fitness function to be optimized. Each particle $i$ in the swarm has a position $x_i(t)$ and a velocity $v_i$ which is updated both by its own and neighbors experience in the search space.

$$x_i(t+1) = x_i(t) + v_i(t+1) \quad (1)$$

The aim is to reach the optimum by updating generations. During this search, each particle updates its position and velocity by a global best solution called $g_{best}$ and a local best solution $p_{best}$. Global best solution is the solution achieved so far whereas local best solution is the best solution found by the particles in that iteration. Having these values, the velocity and the position a particle is obtained as follows;

$$v_i(t+1) = wv_i(t) + c_1 r_1 \left(p_{best,i} - x_i(t)\right) + c_2 r_2 (g_{best} - x_i(t)) \quad (2)$$

where w is the inertia weight factor that aims to adjust the exploration and exploitation abilities of the swarm, $r_1$ and $r_2$ are random numbers between 0 and 1, and $c_1$ and $c_2$ are acceleration coefficients having values between 0 and 4 . The inertia weight factor w in equation (2) can be calculated by using equation (3) where *iter* is the iteration number in a particular instance and $iter_{max}$ is the maximum number of iterations.

$$w = w_{max} - \frac{(w_{max} - w_{min})}{iter_{max}} .iter \quad where \quad 0.4 < w < 0.9 \quad (3)$$

## 2. Controller Tuning (PID Control)

In a feedback (classical) control system (problem), the response of the system is adjusted by the error signal to get the desired response. A controller is generally used to correct the error signal using a control criteria such as a proportional-integral-derivative (PID) controller. Performance of the system can be observed through the overshoot, rise time, settling time, and steady-state error. There are several techniques for tuning controllers; Ziegler_Nichols (Z-N) Oscillation Method, Analytical methods (Pole placement), Frequency response methods and also modern techniques based on Evolutionary Algorithms (EA) and Swarm Intelligence (SI) [6, 7, 8]. These methods have both advantages and disadvantages in use depending on the system characteristics.

The PSO technique was used by researchers in many application areas for controller tuning. Y.Liu et al [9] performed a study on optimization design using PSO Algorithm for PID Controller. Gaing [10] used PSO to optimize the parameters of a PID controller in an AVR system. X. Wang et al [11] used PSO-PID approach for a controller in AQM Routers. Zolkony [12] used PSO for optimization on parameters of power system stabilization. Kao et al [13] used PSO to optimize PID parameters of a slider crank mechanism. Alici et al [14] used the algorithm for prediction of positioning errors of robot manipulators. M. Nasri et al [15] presented a PSO-based optimum design of PID controller for a linear brushless dc motor. Daş et al. [16] applied PSO to control of a four bar mechanism for optimizing PD controllers. Later

Daş et al. [17-18] used this algorithm to a SCARA robot (2 DOF system) and on system identification using PSO-NN approach with an electrohydraulic robot.

## 3. CASE STUDIES

To investigate the performance of the algorithm, three different case studies are presented here. All of the selected systems have practical applications in industry. These systems are available at Dynamic Systems Laboratory of University of Gaziantep at Department of Mechanical Engineering. The first system that is discussed is a one DOF mechanism driven by a brushless DC motor. In order to control this system, the parameters of the PID controller are optimized with PSO algorithm. The second system is a SCARA robot which is used in pick and place type operations. Finally, a hydraulic servo robot, which is a nonlinear system, is dealt in the third case. For this case, a Neural Network trained with PSO algorithm is used for the identification of hydraulic servo robot.

### 3.1. ROBOT CONTROL

#### (I) ONE DOF SYSTEM-LINKAGE AND BLDC MOTOR

In this case, a linkage that is directly driven by a brushless DC servo motor is studied. This configuration is studied together with the actuator dynamics. System simulation is carried out by optimizing PID parameters using the PSO algorithm. The performance of a control system directly depends on tuning of PID parameters $K_p$, $K_v$ and $K_i$ . Optimum controller parameters can give a good step response giving minimum performance criteria. In utilizing the system response with a step input, maximum overshoot $M_p$, rise time $t_r$, settling time $t_s$ and steady state response $e_{ss}$ are applied. The performance criteria in terms of these parameters can be given as:

$$W(K) = (1-e^{-\beta})(M_P + e_{ss}) + e^{-\beta}(t_s - t_r) \qquad (4)$$

where β is given as weight factor and taken as β=0.7 [6]. The fitness function $f$ of the PSO algorithm is defined as the reciprocal of W(K). It is clear from Eq (5) that to obtain a good solution, we have to reach a small W(K) value.

$$f = \frac{1}{W(K)} \qquad (5)$$

The data for electrical motor is taken for a BLDC servo motor [17]. Relevant data for the systems is tabulated in Table 1.

TABLE 1. MECHANICAL AND ELECTRICAL DATA FOR BLDC

| | |
|---|---|
| R=0.80 Ω | L=5.8 mH |
| $K_t$= 0.76 Nm/A | $K_e$=90 V/krpm |
| $J_m$=6.8x10$^{-4}$ kg.m$^2$ | $J_L$=75*$J_m$ kg.m$^2$ |
| D=0.05 Nm.s | |

The PSO algorithm is coded in Matlab. The parameters of the PSO algorithm are selected as follows; population size=80, inertia weight factor $w$, is calculated using Eq (3) where $w_{max}$=0.9 and $w_{min}$=0.4. Acceleration constants are determined as $c_1$=2 and $c_2$=2. PSO reached best PID values after 100 iterations. The values of the controller parameters are $K_p$=114.62, $K_v$=8.03 and $K_i$=76.90. The results can be also seen in Fig.1(a) and (b).

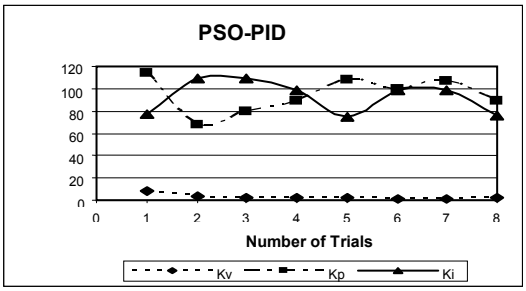



Fig. 1(a). Local Changes of the PID values

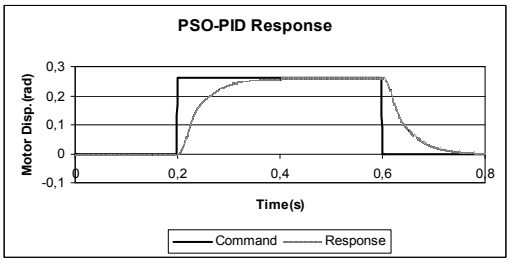



Fig. 1(b).Step response of a BLDC motor

### (II) MOTION CONTROL OF SCARA ROBOT

In this case, the position control of a SCARA robot with 2 axes is studied. The aim is to perform trajectory following using the PSO-PID controller for the RR robot. SCARA is used to pick and place 3 different sized work pieces shown in Fig. 2 (as red-small, blue-medium, green-big). Here the trajectory is taken while the robot performs a picking up operation of a cylinder work piece (of 58 mm diameter and 50 mm length - big one) between two stations (picking-placing point). The motion is performed in 3 seconds. Trajectory data points are recorded during an operation of the robot. Mechanical and electrical data of SCARA is presented in Table 2.



Figure 2. SCARA Robot (Top view)

The following parameters are used to run the PSO algorithm for 1000 iterations; population size = 55, $c_1=c_2=2$, $w_{max}=0.9$ and $w_{min}=0.4$. Using the PSO-PID controller, the following parameters are obtained $K_{p1}=198$, $K_{d1}=102$ and $K_{i1}=51.28$ for the axis 1, the main arm (shoulder) and $K_{p2}=126.62$, $K_{d2}=25.84$ and $K_{i2}=0.83$ for the axis 2, the fore arm (elbow) respectively. Simulation results are shown in Fig. 3(a) and (b) for both axis using PSO-PID approach.

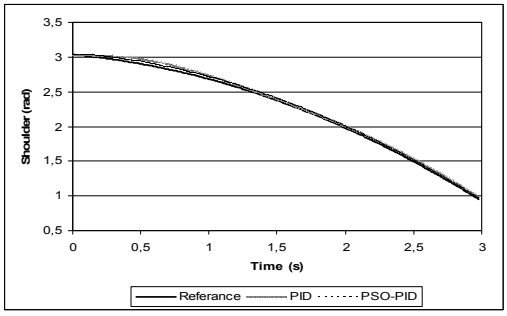



Fig. 3(a). Simulation results for the shoulder axis

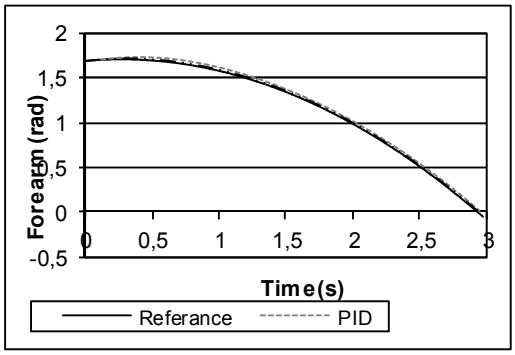



Fig. 3(b). Simulation results for the elbow axis

TABLE 2. MECH. AND ELEC. DATA FOR SCARA

| |
|---|
| $J_1=0.0980$ kg.m$^2$ , $J_2=0.0115$ kg.m$^2$ |
| $m_1=1.90$ kg, $m_2=0.93$ kg |
| $K_{e1}= K_{e2}=0.047$ V/rad/s |
| $K_{t1}= K_{t2}=0.047$ Nm/A |
| $R_{a1}= R_{a2}=3.5\ \Omega$ |
| $L_{a1}= L_{a2}= 1.3$ mH |
| $N_1=90$ , $N_2=220$ (gearbox ratios) |
| Jm= 3.3x10-6 kg.m2, $J_{g1}=0.0002$ kg.m$^2$, $J_{g2}=0.0005$ kg.m$^2$ |

The results presented in Table 3 reveals that PSO-PID produce a better solution compared to the PID approach.

TABLE 3. PERFORMANCE OF SCARA

| ***Positional error (rad)*** | ***PID (Z-N method)*** | ***PSO-PID*** |
|---|---|---|
| The main arm | -0.0728 | -0.0297 |
| The fore arm | -0.0250 | -0.0128 |

### 3.2. ROBOTIC SYSTEM IDENTIFICATION

The system that is considered in this part has two degrees of freedom; as RP configuration. The developed algorithm is applied on the spherical hydraulically driven robot (given in Fig. 4). The first two joints from the ground are revolute joints ($\theta_0$- the base rotation, $\theta_1$- the elevation angle) and the last joint is prismatic ($r_2$ - the reach). The mechanical parameters of the robot are taken from previous studies and shown in Table 4.



Fig. 4. An electrohydraulic robot (RP)

TABLE 4. MECHANICAL PARAMETERS FOR RP ROBOT

| | | | |
|---|---|---|---|
| $\mathbf{R}_0$= 0.0 mm | $R_1$= 280 mm | $S_1$=7 mm | $S_2$=400 mm |
| $L_1$=0.0 mm | $L_2$= 752 mm | | |
| $M_0$=79 kg | $M_1$=49 kg | $M_2$=50 kg | |
| $J_0$=1.19kg.m$^2$ | $J_1$=1.19kg.m$^2$ | $J_2$=1.19kg.m$^2$ | |

Joint 2 (revolute) $-39° \le \theta_1 \le 15°$-
Joint 3 (prismatic) 42.95 cm $\le r_2 \le$ 142.75 cm

The identification of the hydraulic robot is done with a Neural Network trained with a PSO algorithm. Since the problem is identification of a nonlinear system to get better control on the trajectory requirement, a neural network which is good at nonlinear systems is preferred. To develop a NN algorithm firstly 240 data have been recorded experimentally. A circular trajectory is traced by giving the coordinates of a circle to a hydraulic robot, RP configuration.

For the solution of the problem, a NN structure having the input layer N with two nodes (N=2), hidden layer J with twenty nodes (J=20) and output layer L with two nodes (L=2) is used. The input values of the joints ($\theta_2$, $L_2$) are given to the network as input 1 and 2. The output nodes are the responses of the joints. Half of the recorded data have been used for the training purpose and the rest of it for the test. The proposed network is trained with the PSO algorithm (PSO-NN). The objective function of the algorithm is defined as the total error which is difference between desired output value and actual output value. The fitness function of a particle is shown in equation (6).

$$Fitness(particle)_i\, E_t : \frac{\sum (d_{actual} - d_{desired})^2}{2} \quad (6)$$

A population of 45 particles is used for the PSO-NN algorithm. Numbers of particles and hidden layers have been tried on the system in different in various training. The algorithm is run for 5000 iterations. Learning factors are set to $c_1 = c_2 = 2.0$. and the learning rate and the momentum rate is chosen between 0 and 1. Initial weights of the both algorithms are determined between -1 and 1 randomly. During the 5000 iteration, PSO-NN has received the result after 388th iteration in the example. Fig. 5(a), (b) and (c) show the results of identification done by using PSO-NN algorithm.

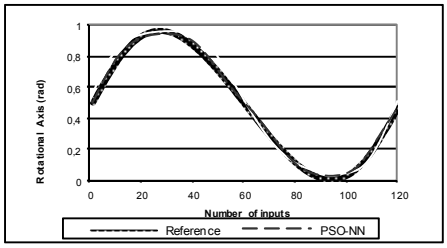



Fig. 5(a) Results for Rotational Axis

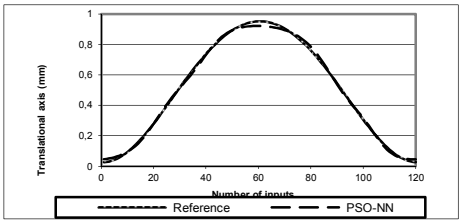



Fig. 5(b) Results for Translational Axis

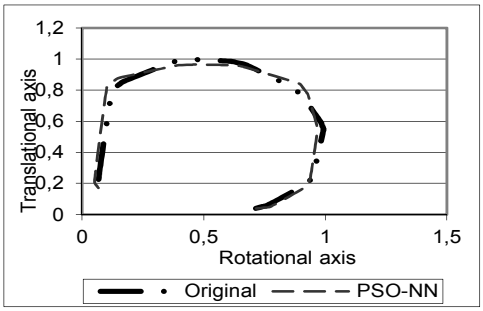



Fig. 5(c). Trajectory Result with PSO-NN

## 4. CONCLUSION

This paper presents case studies dealt with PSO algorithm for different types of controllers in robotic applications. The simulation and experimental results showed that different applications of PSO algorithm - only PSO, PSO-PID and PSO-NN algorithms - has a good in system control ability and training ability. Authors believe that these methods can be applied to other robotic systems for the purpose of control or system identification.

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